Supply Chain Risk Intelligence Dashboard (Power BI)

1. Introduction

This project focuses on analyzing supply chain performance using Power BI. It aims to identify shipment delays, evaluate supplier performance, and highlight high-risk regions affecting operational efficiency.

2. Objective

The objective is to build an interactive dashboard that enables decision-making by providing insights into delays, trends, and risk factors.

3. Data Sources

The project integrates multiple datasets including shipment data, supplier information, weather conditions, and geopolitical risk indicators.

4. Data Modeling

A star schema data model was implemented with shipment data as the fact table and supporting dimension tables such as Date, Country, and Supplier. This structure ensures efficient filtering and accurate analysis.

5. Key Metrics

- Total Shipments
- Delay Percentage
- Average Delay
- High Risk Shipments

These metrics provide a comprehensive overview of supply chain performance.

6. DAX Calculations

Advanced DAX measures were used including:
- Delay % for performance evaluation
- Rolling Delay % for trend smoothing
- CALCULATE for conditional logic

- DATEADD for time comparison
- SELECTEDVALUE for what-if simulation

7. Dashboard Features

- KPI overview section
- Trend analysis charts
- Geographic risk visualization
- Supplier and country ranking
- Drill-through for detailed analysis
- What-if simulation for scenario testing

8. Drill-through Analysis

A drill-through feature allows users to click on a supplier and navigate to a detailed page showing supplier-specific trends and KPIs.

9. What-if Analysis

A parameter-driven approach enables users to simulate changes in delay performance and observe the impact dynamically.

10. Business Impact

This dashboard helps identify bottlenecks, monitor performance, and support strategic decisions to improve supply chain efficiency.

11. Conclusion

The project demonstrates strong skills in data modeling, DAX, visualization, and business analysis using Power BI.